

PH102

After Mid Semester Exams

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Chapter-8

Conservation laws

The Flow of energy

Energy conservation law for electrodynamics

In this chapter we will study the conservation of

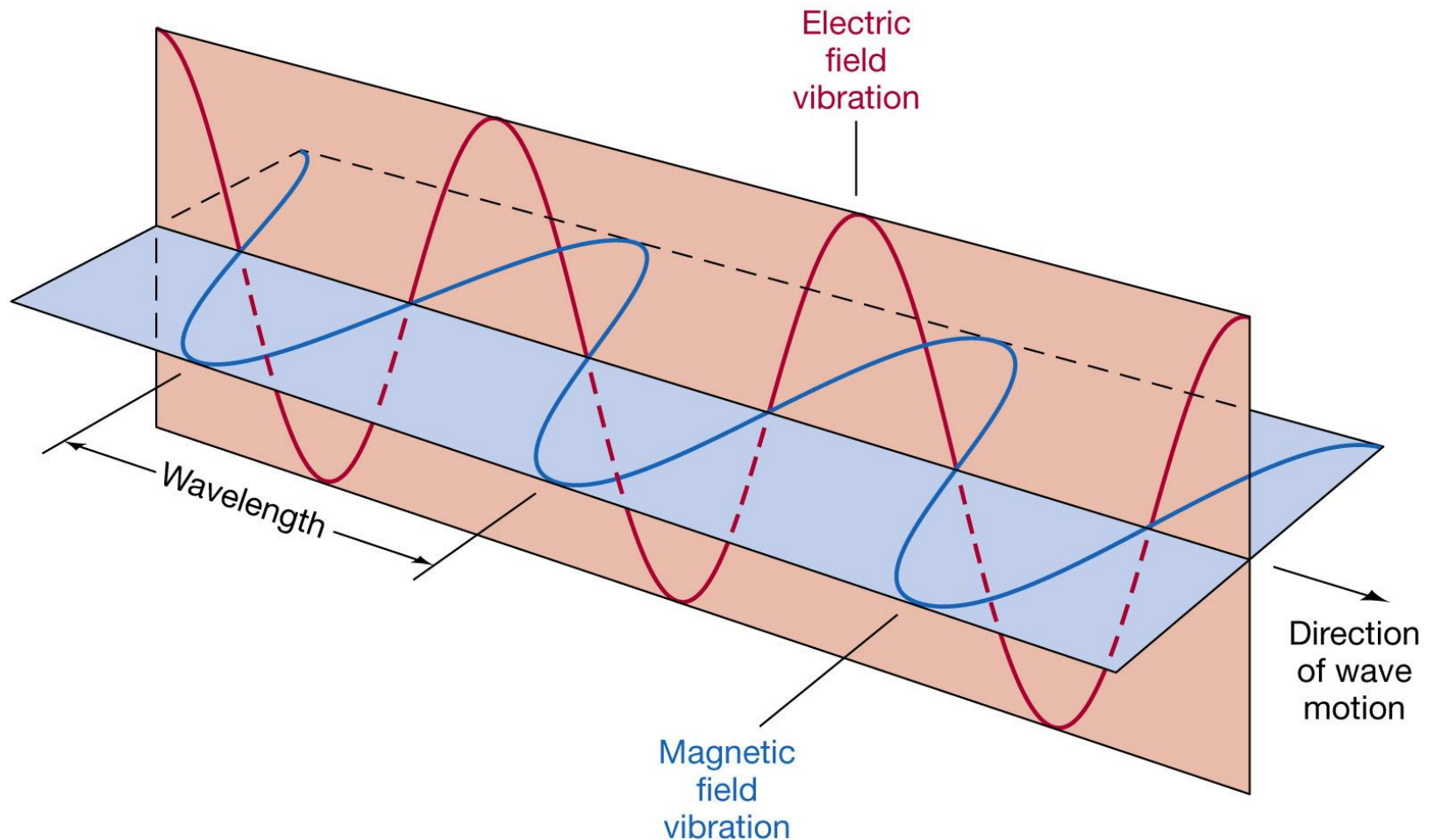
- Energy,
- Poynting's theorem

But lets begin by reviewing the conservation of charge, because it is the paradigm for all conservation laws.

The Flow of energy

The energy per unit time, per unit area, transported by the fields ??

Energy flux density ??



The Continuity Equation

If the total charge in some volume changes, then exactly that amount of charge must have passed in or out through the surface.

The charge in a volume V is

$$Q(t) = \int_V \rho(\vec{r}, t) d\tau$$

The current flowing out through the boundary ' S ' is

$$\int_S \vec{J} \cdot d\vec{a}$$

So local conservation of charge says

$$\frac{dQ}{dt} = - \int_S \vec{J} \cdot d\vec{a}$$

Eq-a

Substituting the value of Q on the left side, and applying the divergence theorem on the right to Eq-a

$$Q(t) = \int_V \rho(\vec{r}, t) d\tau$$

$$\frac{dQ}{dt} = - \int_S \vec{J} \cdot d\vec{a}$$

$$\int_V \frac{\partial \rho}{\partial t} d\tau = - \int_V (\vec{\nabla} \cdot \vec{J}) d\tau$$

Since this is true for any volume so

$$\frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot \vec{J}$$

This is, of course, the continuity equation--the precise mathematical statement of local conservation of charge.

This is the Continuity equation

The purpose of this chapter is to construct the corresponding equations for conservation of energy.

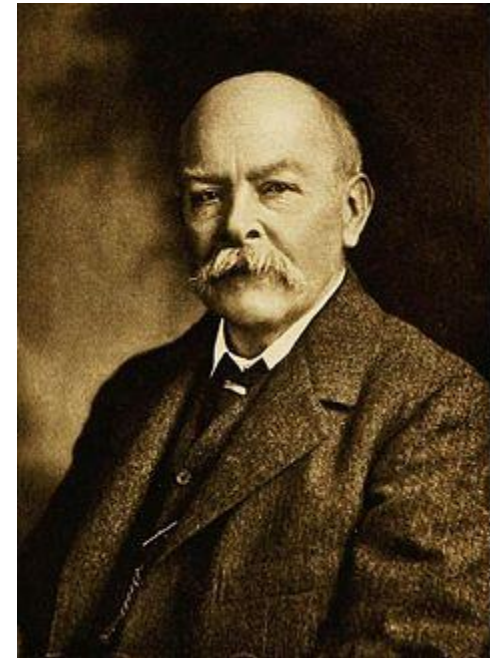
Poynting's Theorem

$$\vec{S} = (\vec{E} \times \vec{H})$$

Analogous to the work-energy theorem in classical mechanics, and mathematically similar to the continuity equation, because it relates the energy stored in the electromagnetic field to the work done on a charge distribution (i.e. an electrically charged object), through energy flux

We know from our previous lectures that the work necessary to assemble a static charge distribution against the Coulomb repulsion of like charges is,

$$W_e = \frac{\epsilon_0}{2} \int E^2 d\tau$$



John Henry Poynting

1852 –1914

was an English physicist

Similarly, the work required to get currents going (against the back emf is),

$$W_m = \frac{1}{2\mu_0} \int B^2 d\tau$$

Hence, the total energy stored in electromagnetic fields is

$$U_{em} = \frac{1}{2} \int (\epsilon_0 E^2 + \frac{B^2}{\mu_0}) d\tau$$

Equation 1

Lets prove the above equation in more generally, now, in the context of the energy conservation law for electrodynamics.

The energy conservation law for electrodynamics.

Suppose we have some charge and current configuration which, at time t , produces fields $\mathbf{E}$ and $\mathbf{B}$. In the next instant, dt , the charges move around a bit.

Question: How much work, dW , is done by the electromagnetic forces acting on these charges in the interval dt ?

Solution: According to the Lorentz force law, the work done on a charge q

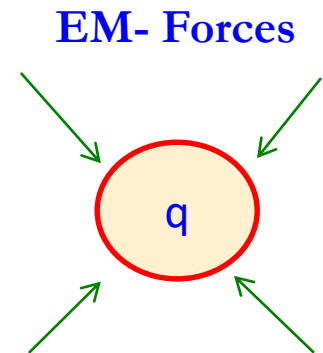
$$\vec{F} \cdot d\vec{l} = q \left(\vec{E} + (\vec{v} \times \vec{B}) \right) \cdot \vec{v} dt$$

$$\vec{F} \cdot d\vec{l} = q \vec{E} \cdot \vec{v} dt$$

$$q = \rho d\tau$$
$$\mathbf{J} = \rho \mathbf{v}$$

And

$$dW = \vec{F} \cdot d\vec{l} = q \vec{E} \cdot \vec{v} dt = \rho d\tau \vec{E} \cdot \vec{v} dt$$



Since $(\vec{v} \times \vec{B}) \cdot \vec{v} dt = 0$

$$dW = \vec{F} \cdot d\vec{l} = q\vec{E} \cdot \vec{v} dt = \rho d\tau \vec{E} \cdot \vec{v} dt$$

$$dW = d\tau \vec{E} \cdot \rho \vec{v} dt$$

$$\rho v = \vec{j}$$

$$dW = d\tau \vec{E} \cdot \vec{j} dt$$

Hence, the rate at which work is done on all the charges in a volume 'V' is:

$$\frac{dW}{dt} = \int_V \vec{E} \cdot \vec{j} d\tau$$

Equation 2

Evidently $\vec{E} \cdot \vec{j}$ is the work done per unit time, per unit volume; which is to say, the power delivered per unit volume.

We can express this quantity in terms of the fields alone. Using the Ampere-Maxwell law to eliminate $\vec{J}$.

$$\vec{E} \cdot \vec{J} = \frac{1}{\mu_0} \vec{E} \cdot (\nabla \times \vec{B}) - \varepsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

From the relation

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$

$$\nabla \cdot (\vec{E} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{B})$$

Use the Faraday's law

Using the Faraday's law

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot (\vec{E} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{B})$$

$$\vec{E} \cdot (\nabla \times \vec{B}) = -\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} - \nabla \cdot (\vec{E} \times \vec{B})$$

Also, we can write

$$\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (B)^2$$

$$\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (E)^2$$

$$\vec{E} \cdot \vec{j} = \frac{1}{\mu_0} \vec{E} \cdot (\nabla \times \vec{B}) - \varepsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

Hence

$$\vec{E} \cdot \vec{j} = -\frac{1}{2} \frac{\partial}{\partial t} \left(\epsilon_0 \vec{E}^2 + \frac{1}{\mu_0} \vec{B}^2 \right) - \frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B})$$

Substituting this into Equation 2, and applying the divergence theorem to the second term, we have

$$\frac{dW}{dt} = \int_V \vec{E} \cdot \vec{j} d\tau$$

$$\frac{dW}{dt} = -\frac{\partial}{\partial t} \int_V \frac{1}{2} \left(\epsilon_0 \vec{E}^2 + \frac{1}{\mu_0} \vec{B}^2 \right) d\tau - \frac{1}{\mu_0} \oint_S (\vec{E} \times \vec{B}) \cdot d\vec{a}$$

This is Poynting's theorem

Here 's' is the surface bounding 'V'.

It is the "**work-energy theorem**" of electrodynamics.

The first integral on the right is the total energy stored in the fields, U_{em} (Equation 2). The second term evidently represents the rate at which energy is carried out of V, across its boundary surface, by the electromagnetic fields.

Poynting's theorem says, then, that the work done on the charges by the electromagnetic force is equal to the decrease in energy stored in the field less the energy that flowed out through the surface.

The energy per unit time, per unit area, transported by the fields is called the **Poynting vector**:

Poynting vector



$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$\vec{S} = (\vec{E} \times \vec{H})$$

Specifically, $\vec{S} \cdot \overrightarrow{da}$ is the energy per unit time crossing the infinitesimal surface da the energy flux, if you like (so $\mathbf{S}$ is the energy flux density).

$$\frac{dW}{dt} = -\frac{dU_{em}}{dt} - \oint_S \vec{S} \cdot \overrightarrow{da}$$

The work W done on the charges will increase their mechanical energy (kinetic, potential, or whatever). If we let u_{mech} denote the mechanical energy density, so that

$$\frac{dW}{dt} = \frac{d}{dt} \int_V u_{\text{mech}} d\tau$$

u_{em} for the energy density of the fields

$$u_{\text{em}} = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$$

$$\therefore -\frac{\partial}{\partial t} \int_V \frac{1}{2} \left(\epsilon_0 \vec{E}^2 + \frac{1}{\mu_0} \vec{B}^2 \right) d\tau - \frac{1}{\mu_0} \oint_S (\vec{E} \times \vec{B}) \cdot d\vec{a} = 0$$

Then

$$\frac{d}{dt} \int_V (u_{\text{mech}} + u_{\text{em}}) d\tau = - \oint_S \vec{S} \cdot d\vec{a}$$

$$= - \int_V (\nabla \cdot \vec{S}) d\tau$$

Hence

$$\frac{\partial}{\partial t} (u_{mech} + u_{em}) = -\nabla \cdot \vec{S}$$

This is the differential version of Poynting's theorem.

Comparing this equation with Continuity equation

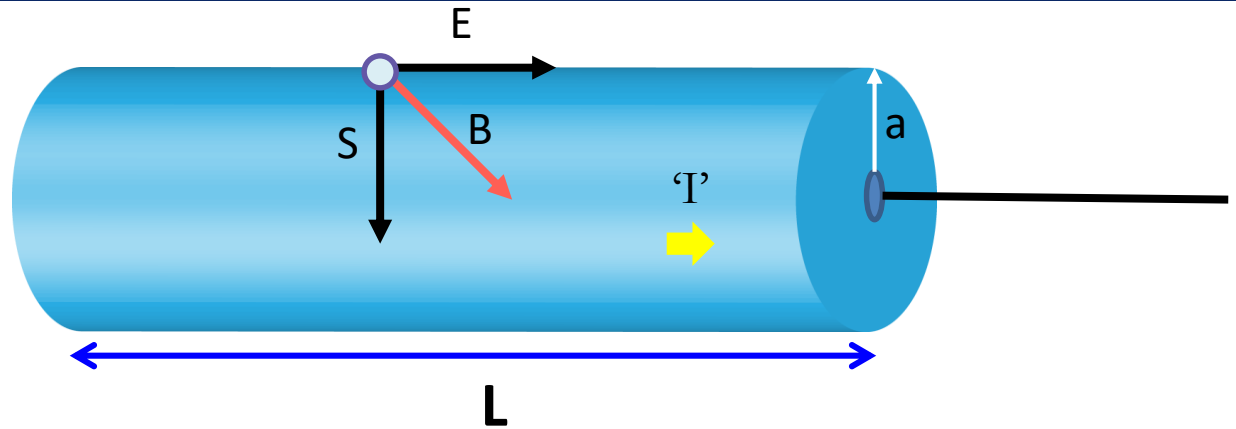
$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \vec{j}$$

The charge density is replaced by the energy density (mechanical plus electromagnetic), and the current density is replaced by the Poynting vector.

Physical Interpretation: The flow of energy in exactly the same way that $\vec{j}$ describes the flow of charge.

Example 1: Consider 'I' current flows down a wire. Find the energy per unit time delivered to the wire using Poynting's Theorem and also compare the result with the Joule's heating law? Consider V = potential difference between the two ends of the cylinder.

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$



Solution: The electric field parallel to the wire is

$$E = \frac{V}{L}$$

Here, V is the potential difference between the ends and L is the length of the wire

The magnetic field is "circumferential"; at the surface (radius a) it has the value

$$B = \frac{\mu_0 I}{2\pi a}$$

Accordingly, the magnitude of the Poynting vector is

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$S = \frac{1}{\mu_0} \left(\frac{\mu_0 I}{2\pi a} \frac{V}{L} \right)$$

$$S = \frac{VI}{2\pi La}$$

It points radially inward.

The energy per unit time passing in, through the surface of the wire is therefore

$$\int \vec{S} \cdot d\vec{a} = S(2\pi La) = \frac{VI}{2\pi La} (2\pi La)$$

$$\int \vec{S} \cdot d\vec{a} = VI$$

$$P = VI$$

Joule's law can be stated as

Energy dissipated per unit time =
Energy dissipated per charge passing through resistor $\times$ Charge passing
through resistor per unit time

Our result obtained from the Poynting's Theorem is exactly same as by
Joule's law.

Example 2: Calculate the power (energy per unit time) transported along coaxial cable carries current $\mathbf{I}$ (the current flows down the surface of the inner cylinder, radius $\mathbf{a}$, and back along the outer cylinder, radius $\mathbf{b}$) as shown in Figure, assuming the two conductors are held at potential difference $\mathbf{V}$, and carry current $\mathbf{I}$ (down one and back up the other).

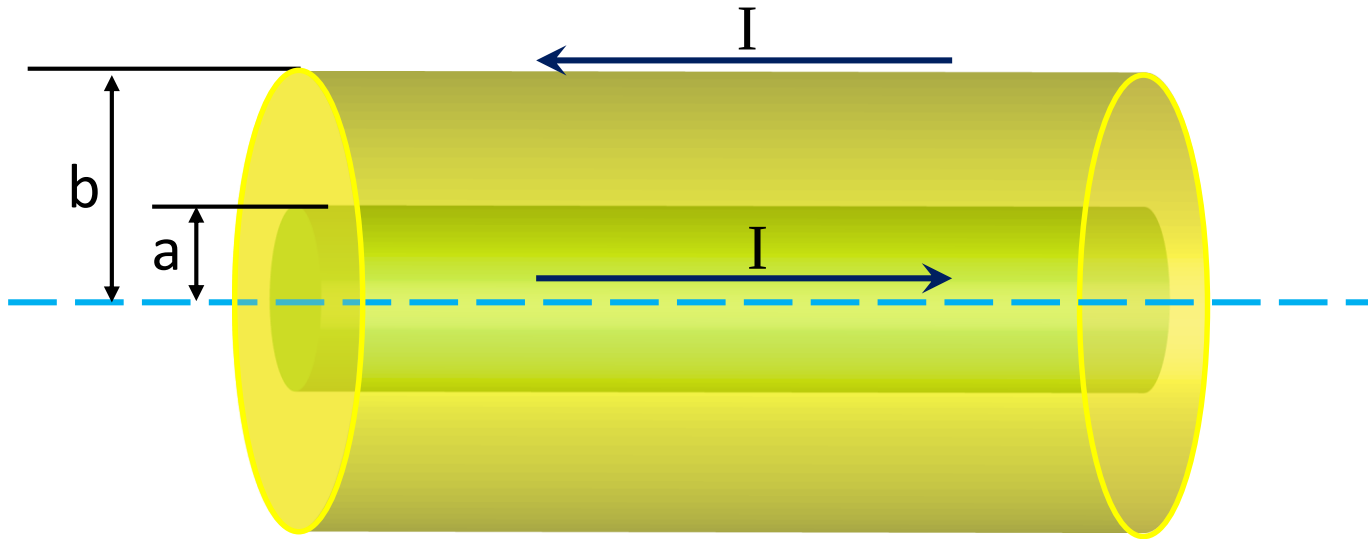


Figure 1

Solution: As we know from our previous lecture's example

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0} \frac{1}{s} \hat{s}$$

$$\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

Using, the Poynting vector theorem

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$\vec{S} = \frac{\lambda I}{4\pi^2 \epsilon_0} \frac{1}{s^2} \hat{z}$$

$$\because \hat{s} \times \hat{\phi} = \hat{z}$$

$$P = \int \vec{S} \cdot \overrightarrow{d\vec{a}}$$

$$P = \int_a^b S 2\pi s \, ds$$

$$S = \frac{\lambda I}{4\pi^2 \epsilon_0} \frac{1}{s^2}$$

$$P = \int_a^b \frac{\lambda I}{4\pi^2 \epsilon_0} \frac{1}{s^2} 2\pi s \, ds$$

$$P = \frac{\lambda I}{2\pi \epsilon_0} \int_a^b \frac{1}{s} ds$$

$$P = \frac{I\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

But

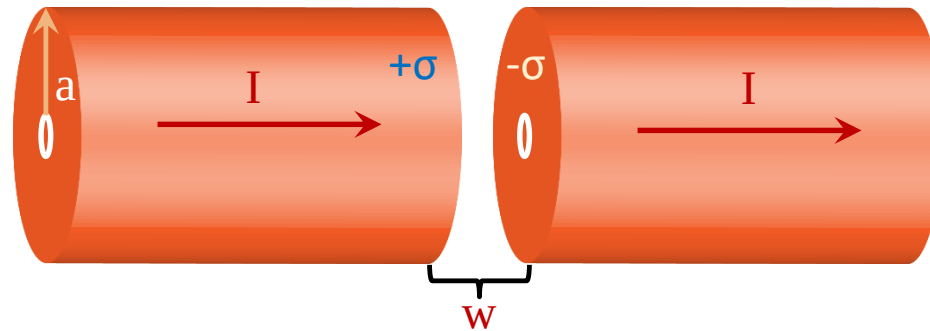
$$V = \int_a^b \vec{E} \cdot d\vec{l}$$

$$V = \frac{\lambda}{2\pi\epsilon_0} \int_a^b \frac{1}{s} ds \quad \Rightarrow \quad V = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

Hence

$$P = IV$$

Example: A fat wire, radius a , carries a constant current I , uniformly distributed over its cross section. A narrow gap in the wire, of width $w \ll a$, forms a parallel-plate capacitor, as shown in Figure .

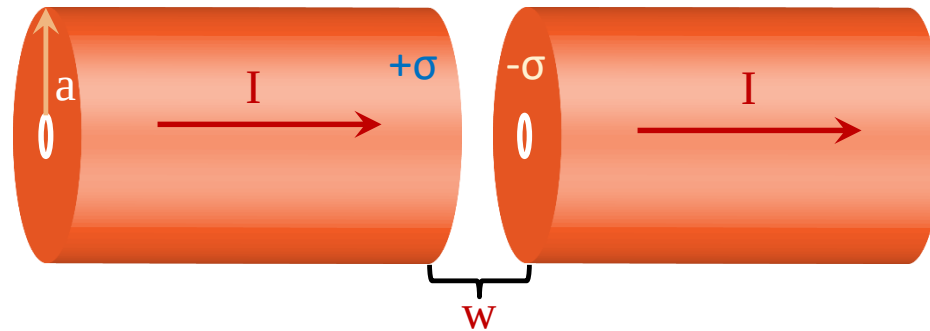


Questions:

- Find the electric and magnetic fields in the gap, as functions of the distance s from the axis and the time t . (Assume the charge is zero at $t = 0$.)
- Find the energy density u_{em} and the Poynting vector $\mathbf{S}$ in the gap. Note especially the direction of $\mathbf{S}$.

Check that the Equation: $\frac{\partial}{\partial t} (u_{mech} + u_{em}) = -\nabla \cdot \vec{S}$ is satisfied.

Example: A fat wire, radius a , carries a constant current I , uniformly distributed over its cross section. A narrow gap in the wire, of width $w \ll a$, forms a parallel-plate capacitor, as shown in Figure .



Question:

c) Determine the total energy in the gap, as a function of time. Calculate the total power flowing into the gap, by integrating the Poynting vector over the appropriate surface. Check that the power input is equal to the rate of increase of energy in the gap.

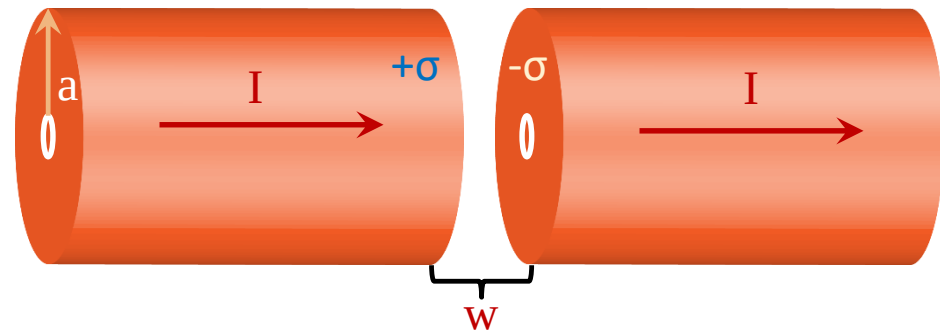
a) Find the electric and magnetic fields in the gap, distance s from the axis and the time t . (Assume the charge is zero at $t = 0$.)

Solution: Part (a)

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{z}$$

$$\sigma = \frac{Q}{\pi a^2}$$

$$Q(t) = It$$



Hence the Electric Field

$$\vec{E}(t) = \frac{It}{\pi a^2 \epsilon_0} \hat{z}$$

Equation A

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

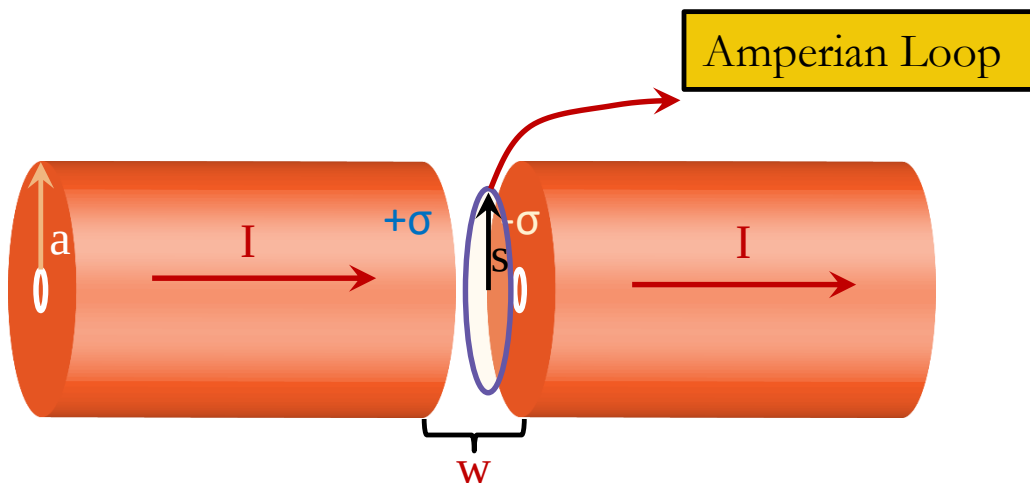
$$\vec{J}_d = \epsilon_0 \frac{\vec{I}}{\pi a^2 \epsilon_0}$$

From Equation 6

$$\vec{E}(t) = \frac{It}{\pi a^2 \epsilon_0} \hat{z}$$

$$\vec{J}_d = \frac{I}{\pi a^2} \hat{z}$$

Drawing an “Amperian loop” at radius ‘s’,



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{d-enclosed}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{d-enclosed}$$

$$B \times 2\pi s = \mu_0 I_{d-enclosed}$$

$$= \mu_0 \frac{I}{\pi a^2} \pi s^2$$

$$= \mu_0 \frac{Is^2}{a^2}$$

$$B \times 2\pi s = \mu_0 \frac{Is^2}{a^2}$$

Hence the Magnetic Field

$$B(s, t) = \frac{\mu_0 Is}{2\pi a^2} \hat{\phi}$$

Part (b): Our aim is to check the validity of the equation:

$$\frac{\partial}{\partial t} (u_{mech} + u_{em}) = -\nabla \cdot \vec{S}$$

L.H.S

R.H.S

Considering the L.H.S

Here only the electro-magnetic contribution is present $u_{mech} = 0$

$$u_{em} = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$$

$$u_{em} = \frac{1}{2} \left(\epsilon_0 \left(\frac{It}{\pi a^2 \epsilon_0} \right)^2 + \frac{1}{\mu_0} \left(\frac{\mu_0 Is}{2\pi a^2} \right)^2 \right)$$

$$u_{em} = \frac{1}{2} \left(\epsilon_0 \frac{I^2 t^2}{\pi^2 a^4 \epsilon_0^2} + \frac{1}{\mu_0} \frac{\mu_0^2 I^2 s^2}{4\pi^2 a^4} \right)$$

$$u_{em} = \frac{1}{2} \left(\frac{I^2 t^2}{\pi^2 a^4 \epsilon_0} + \frac{\mu_0 I^2 s^2}{4\pi^2 a^4} \right)$$

$$u_{em} = \frac{1}{2} \left[\frac{\mu_0 I^2}{\pi^2 a^4} \right] \left(\frac{t^2}{\mu_0 \epsilon_0} + \frac{s^2}{4} \right)$$

Since velocity of light 'c' is equal to



$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$\epsilon_0 \approx 8.85 \times 10^{-12} \text{ F}\cdot\text{m}^{-1}$ (or $\text{A}^2\cdot\text{s}^4\cdot\text{kg}^{-1}\cdot\text{m}^{-3}$ in [SI base units](#), or $\text{C}^2\cdot\text{N}^{-1}\cdot\text{m}^{-2}$ or $\text{C}\cdot\text{V}^{-1}\cdot\text{m}^{-1}$ using other SI coherent units)

$\mu_0 = 4\pi \times 10^{-7} \text{ H}\cdot\text{m}^{-1} \approx 1.256 \times 10^{-6} \text{ H}\cdot\text{m}^{-1}$ or $\text{N}\cdot\text{A}^{-2}$

$$u_{em} = \frac{1}{2} \left[\frac{\mu_0 I^2}{\pi^2 a^4} \right] \left((ct)^2 + \left(\frac{s}{2} \right)^2 \right)$$

$$\frac{\partial}{\partial t} (u_{em}) = \frac{I^2 t}{\pi^2 a^4 \epsilon_0}$$

$$\frac{\partial}{\partial t} (u_{mech} + u_{em}) = -\nabla \cdot \vec{S}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$\vec{S} = \frac{1}{\mu_0} \left(\frac{It}{\pi a^2 \epsilon_0} \hat{z} \times \frac{\mu_0 Is}{2\pi a^2} \hat{\phi} \right)$$

$$\vec{S} = -\frac{I^2 t s}{2\pi^2 \epsilon_0 a^4} \hat{s}$$

$$\hat{z} \times \hat{\phi} = -\hat{s}$$

$$-\nabla \cdot \vec{S} = \frac{I^2 t}{2\pi^2 \epsilon_0 a^4} (\nabla \cdot (s \hat{s}))$$

Since

$$\nabla \cdot \vec{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

$$\begin{aligned}\nabla \cdot (s\hat{s}) &= \frac{1}{s} \frac{\partial}{\partial s} (s^2) \\ &= \frac{1}{s} 2s\end{aligned}$$

$$\nabla \cdot (s\hat{s}) = 2$$

$$-\nabla \cdot \vec{S} = \frac{I^2 t}{2\pi^2 \epsilon_0 a^4} (\nabla \cdot (s\hat{s}))$$

$$\frac{\partial}{\partial t} (u_{em}) = \frac{I^2 t}{\pi^2 a^4 \epsilon_0} = -\nabla \cdot \vec{S} = \frac{I^2 t}{\pi^2 \epsilon_0 a^4}$$

Hence

$$\frac{\partial}{\partial t} (u_{mech} + u_{em}) = -\nabla \cdot \vec{S}$$